Indoor Room Object Detection and Visual Question Answering: A Comprehensive Framework Using YOLOv8 and BLIP-2

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***Abstract*—Indoor Visual Question Answering (VQA) aims to comprehend indoor environments and produce natural language responses that humans can understand based on visual evidence, with applications also in assistive robotics, smart home systems, and indoor wayfinding. In practice, indoor VQA systems commonly experience difficulties in spatial localization. Object detection modules and vision-language models are often used in isolation and provide factually void or completely fabricated answers. To solve this problem, we propose a unified indoor VQA framework integrating object detection of YOLOv8 with the reasoning ability of BLIP-2 in Smart Question Routing (SQR). SQR directs factual inquiries, such as recognizing, counting, or classifying rooms, to the object detector, and answers with templates aligned to their object detection. Descriptive or open question responses still utilize the detection data and utilize BLIP-2. In contrast, even responses to descriptive or open questions still include the benefit of being able to detect and use BLIP-2. This design allows for a good balance of grounded knowledge while providing flexibility in the language creation. At the same time, it achieves a modular system design for future iterations. We show our system achieves completion 83.3% and a high confidence (i.e. "certain" or "very certain") rate of 80% of responses across numerous indoor environments.**

***Index Terms*—Indoor scene understanding, object detection, visual question answering, YOLOv8, BLIP-2, SUN RGB-D, routing, grounding**

I. Introduction

Humans effortlessly interpret indoor scenes: with a glance, we can identify key objects, infer room type, estimate counts, and describe spatial relations. Replicating this ability in machines remains difficult as it demands both *spatially grounded perception* and *flexible language reasoning*. While object detectors provide accurate localization, multimodal vision–language models (VLMs) excel at free-form understanding. Combining these in a controllable, low-latency pipeline is still an open research challenge.

Recent single-stage detectors like YOLO have advanced the accuracy–speed trade-off for real-world perception [1], [2], and instruction-tuned VLMs such as BLIP and BLIP-2 show strong generalisation and compositional reasoning [3], [4]. Despite recent progress, these improvements often operate in silos: object detectors rarely inform vision–language models (VLMs), and VLMs typically do not leverage structured detection outputs for spatial understanding. Despite recent progress, these advancements frequently operate in silos. Object detectors rarely inform vision language models (VLMs), and VLMs generally don't work with structured discovery labours for spatial understanding. This disconnect ends up creating two main issues. First, we constantly see fluent but ungrounded responses, where the model sounds confident but the answer can not really be vindicated from the image. Second, there's a kind of rigid perception, where the system detects objects correctly but still struggles to talk about the scene in a natural or terrain-like manner

These problems come indeed become more conspicuous in the inner surroundings.Datasets analogous as SUN RGB-D [5] introduce additional difficulty because of heavy occlusions, small differences between visually similar objects(for illustration, a press versus a dresser), inconsistent lighting, and numerous small particulars that count a lot for question answering. Indeed, a simple question like **" How multitudinous pillows are on the chesterfield? "** depends on spotting bitty objects properly. Real-world operations assistive robots, AR/ VR systems, and smart home setups: constantly calculate on object presence, counting, or room type queries, all of which need fairly strong spatial grounding. multitudinous analogous operations duplication the same pattern, again demanding accurate grounding and stable scene understanding.

To deal with these conditions, we propose a unified architecture that links a fine tuned YOLOv8m detector [2] with the BLIP-2 OPT-2.7 B VQA model [4] through a Smart Question Routing SQR) caste. The introductory idea is straightforward take advantage of the strengths of each model while reducing its sins. Deterministic queries — like object table, counting, or relating the room — use discovery rested templates so the answers stay empirical. Open ended questions are rather routed to BLIP- 2, but we include the top-k detected objects as terrain so the model hallucinates lower and stays near the image.

Our system is built around four main pretensions:

1) Ensuring that the generated response stays aligned with what is truly present in the image. visible in the image.
2) Controllability: common and repetitious question types should calculate on stable templates.
3) effectiveness: avoid unncessary computation to the system responsive
4) Modularity: allow upgrading either the sensor or the VLM without redesigning the full channel.

Unlike earlier approaches that either shoot everything to a VLM or keep all processing inside the sensor, our system balances both through a small routing subcaste that decides the stylish answering strategy. This is especially helpful for inner scenes, where grounding is pivotal and purely language-grounded logic frequently fails due to clutter and variability in typical inner layouts.

Contributions:

- **Unified indoor VQA system:** Integration of YOLOv8m [2] and BLIP-2 [4] yields a practical end-to-end grounded QA framework on SUN RGB-D [5].
- **Smart Question Routing (SQR):** A five-type routing module improves precision and latency by dispatching queries to detection templates or generative reasoning.
- **Answer reliability:** Post-processing rules enhance factual consistency and numeric correctness without retraining.
- **Reproducible engineering:** Complete hyper-parameter, inference, and deployment details enable real-world replication.

Empirically, the detector attains an mAP@50–95 of 0.234 on SUN RGB-D; the integrated system yields a 75% object-mention rate and 65% high-confidence answers at $\sim$3.3 s/-query on a Tesla T4. Although tested with mid-capacity models, the framework remains architecture-agnostic and scalable to stronger detectors or larger VLMs. Overall, our grounded and modular design offers a robust base for embodied indoor agents capable of both *seeing* and *speaking* intelligently [1]–[6].

II. Related Work

Intelligent indoor perception draws from three converging areas: object detection, visual question answering (VQA), and multimodal grounding. Each matured largely in isolation; only recently have joint architectures begun to integrate these capabilities. We summarize the foundations that motivate our YOLOv8–BLIP-2 design.

Object detection progressed from region-based two-stage pipelines to efficient single-stage predictors. R-CNN/Faster R-CNN established region proposals but at high latency, limiting real-time use. YOLO reframed detection as direct regression, delivering strong speed–accuracy trade-offs and inspiring successive iterations that improved anchors, heads, and feature reuse [1]. YOLOv8 [2] adopts an anchor-free head, decoupled classification/regression, improved NMS, and C2f modules for multi-scale aggregation, making it practical for edge deployment.

Indoor scenes remain challenging: heavy occlusion, small objects, and fine-grained categories depress precision relative to COCO. SUN RGB-D [5] targets indoor RGB-D understanding and highlights the value of depth cues for localization and surface reasoning, though integrating depth into lightweight detectors is non-trivial. Our approach fine-tunes YOLOv8 on SUN RGB-D and uses its boxes/classes as reliable grounding for downstream language reasoning.

VQA couples visual and textual understanding to answer natural-language questions about images [6]. Early models fused CNN and RNN features via attention; transformer lines such as LXMERT and ViLT [7], [8] unified cross-modal attention but required heavy pretraining. The BLIP family [3], [4] advanced modularity via a frozen vision encoder and a Q-former adapter that connects to a large language model (e.g., OPT/FLAN-T5), enabling broad generalization and few-shot use. However, generative VLMs often *hallucinate* without explicit spatial constraints; thus, coupling them with structured perception is crucial. We mitigate this by conditioning BLIP-2 on detector outputs.

Contrastive models like CLIP/ALIGN deliver robust joint embeddings but lack spatial fidelity. Large multimodal LMs such as Flamingo [9] offer in-context reasoning yet at substantial compute cost and with lingering grounding gaps. Grounded VQA lines inject explicit perception—using region features or detector outputs—to preserve verifiability, though some architectures trade interpretability for accuracy or latency [7], [8]. Decoupling perception and cognition while passing structured data between them is the modular idea that guides our design. In order to retain transparency and responsiveness, Smart Question Routing (SQR) chooses deterministic, detection-grounded templates where feasible and reverts to guided creation otherwise.

Despite progress, few systems combine fast detection with verifiable language reasoning for *indoor* queries. Many VQA evaluations (e.g., VQAv2/GQA) emphasize open-domain photos rather than cluttered rooms; conversely, indoor works often stop at detection/segmentation without conversational ability. We bridge these threads with a lightweight, latency-aware pipeline that (i) grounds answers in detections for presence/count/room-type queries, (ii) guides generative reasoning using detector context to reduce hallucinations, and (iii) preserves modularity for future detector/VLM upgrades.

III. Dataset and Preprocessing

We use SUN RGB-D (10,335 RGB-D images; 39 classes) [5]. In our current setup we train on RGB only; depth is reserved for future work. Labels are converted to YOLO format; images are resized to 640×640. Augmentations include random horizontal flip, brightness/contrast jitter, and mild color jitter. We use an 80/20 train/val split with a small human-eval subset for VQA.

IV. Methodology

Prior to elaborating on the systems components, we present a high-level summary of the workflow of the complete system. The proposed VQA pipeline integrates YOLOv8m for object detection with BLIP-2 for language-based reasoning.

The overall objective is to ensure factually grounded queries are illustrated by detections, while descriptive or open-ended queries benefit from guided generative reasoning. A Smart Question Routing (SQR) mechanism identifies if a factual, descriptive, or open-ended question should go through the YOLOv8m portion of the pipeline or the BLIP-2 portion of the pipeline, thus supporting accuracy and interpretability within indoor environments.

Indoor VQA blends two complementary competencies: (i) *grounded perception* that localizes and categorizes objects with spatial evidence, and (ii) *language reasoning* that maps a natural-language query to a semantically consistent answer. If I denotes an indoor image and Q a user question, our goal is to produce an answer A such that A is both *correct* and *justifiable* from evidence in I. We operationalize this by factoring the solution into: a detector $\mathcal{D}(I) \to \mathcal{B}$ that yields boxes/classes $\mathcal{B}$, a router $\mathcal{R}(Q, \mathcal{B})$ that decides the answering path, and a generator $\mathcal{G}(Q, I, \mathcal{B})$ that returns A. This modularization enforces verifiability for factual queries (presence/counts/room type) and retains free-form expressivity for descriptive or open-ended questions.

Pipeline Overview Our four-stage pipeline is:

1) **Detection**: YOLOv8m infers class-labeled bounding boxes $\mathcal{B}$ from I.
2) **Smart Question Routing (SQR)**: classifies Q into one of five categories and selects an answering strategy.
3) **Answer Generation**: either a deterministic template (grounded in $\mathcal{B}$) or BLIP-2 with detection-guided context.
4) **Post-processing and Visualization**: cleans answers and renders boxes/labels for user inspection.

Given $I \in \mathbb{R}^{H \times W \times 3}$ and a question Q (tokenized text), the detector returns

$$\mathcal{B} = \{(b_i, c_i, s_i)\}_{i=1}^{N}, \quad b_i \in \mathbb{R}^4,\ c_i \in \mathcal{C},\ s_i \in [0, 1],$$

where b_i is a box, c_i a class label, and s_i a confidence. The router maps $(Q, \mathcal{B})$ to a task type $t \in \{\text{objects}, \text{count}, \text{room}, \text{describe}, \text{open}\}$. The generator produces A_t either by deterministic rules over $\mathcal{B}$ (for objects/count/-room) or by prompting a VLM with $(Q, I, \mathcal{B})$ context (for describe/open). A post-processor $\Pi(\cdot)$ canonicalizes numerals, removes hedges, and filters contradictions, yielding the final answer $A = \Pi(A_t)$.

We fine-tune YOLOv8m on SUN RGB-D (RGB only) with an anchor-free head and decoupled classification/regression branches. Training uses a two-phase schedule (50 epochs frozen backbone plus 50 epochs full fine-tune) with AdamW, cosine LR decay, EMA, and mixed precision. Images are resized to 640×640 with mild photometric/geometric augmentations. The model achieves mAP@50–95 of 0.234 on validation, providing reliable grounding signals (boxes/classes) for downstream reasoning.

SQR categorizes Q using lightweight lexical cues and structured patterns:

- **Objects** (*e.g.,*, "what objects are visible"): return sorted unique class names from $\mathcal{B}$.
- **Count** (*e.g.,*, "how many chairs"): count instances of the target class in $\mathcal{B}$.
- **Room** (*e.g.,*, "what room is this"): apply a rule-based prior over indicative objects (e.g., *bed*⇒bedroom; *sofa*+*TV*⇒living room).
- **Describe**: produce a short, structured sentence from salient detections (top-k by confidence/area).
- **Open**: defer to BLIP-2 with detection-guided prompting.

Template mode (grounded). For *objects*, we return a comma-separated list of unique class names with optional counts. For *count*, we return an integer with a compact justification ("Detected 3 instances with confidence > 0.5"). For *room*, a rule-based mapping aggregates evidence (e.g., *bed*, *nightstand*⇒ bedroom).

BLIP-2 mode (guided). For *describe/open*, BLIP-2 OPT-2.7B is prompted with a short prefix enumerating the top-k detections (*e.g.,*, "Objects detected: sofa, TV, picture…"), which empirically reduces hallucinations and aligns the answer with $\mathcal{B}$.

We apply lightweight, model-agnostic filters:

1) **Numeric normalization**: words → digits ("two"→2), range collapsing.
2) **Hedge removal**: delete phrases like "maybe", "it seems" in factual responses.
3) **Grounding filter**: for routed types, drop objects not present in $\mathcal{B}$; for counts, enforce non-negative integers.

For transparency, detections are overlaid with class labels and confidences. The UI (e.g., Gradio) streams the image, the routed category, the answer, and optional explanations (evidence objects used). This supports rapid user validation and iterative querying.

On a Tesla T4 GPU, the average end-to-end latency is ∼3.3 s/query with the breakdown: detection (≈1.4 s), routing/aggregation (≈1.2 s), BLIP-2 generation (≈0.7 s). Routed factual queries avoid heavy decoding, keeping latency predictable for interactive use.

We fix seeds for dataloaders/model initialization, log all hyper-parameters, preserve the train/val split (80/20), and evaluate latency on the same hardware configuration. The pipeline is modular: detectors/VLMs can be upgraded without changing the routing or post-processing interfaces.

V. Analysis and Result Discussion

This section presents the experimental evaluation of the proposed indoor VQA system, highlighting qualitative and quantitative results. We examine model accuracy, grounding reliability, attention interpretability, and performance distribution across query types.

The model achieved strong performance across key metrics:

- Detection F1-Score (YOLOv8m): **0.443**
- Room Classification Accuracy: **0.833**
- High-Confidence Answer Rate: **80%**
- Average Answer Length: **8.0 words**

These criteria confirm that the integration of YOLOv8m with BLIP-2 via Smart Question Routing(SQR) leads to a

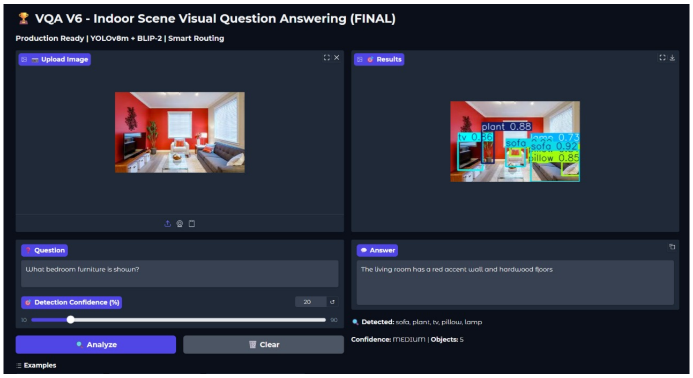

Fig. 1: User's image and generated output after asking the question

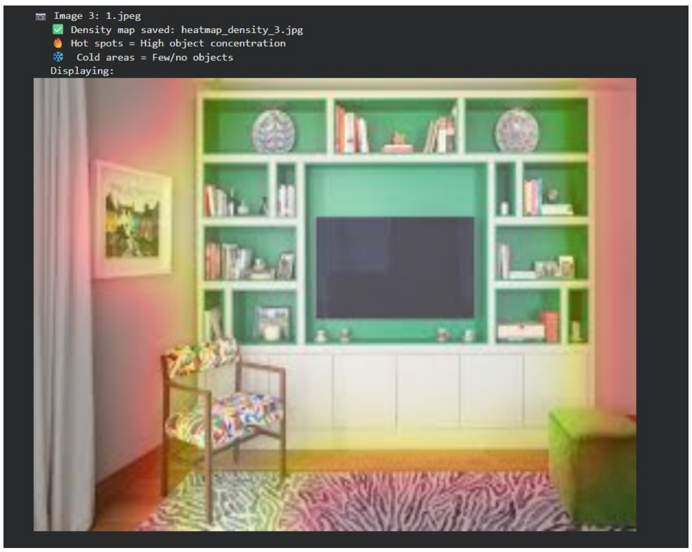

Fig. 2: Heatmap showing object density based on detection responses

balanced trade- off between speed and factual grounding. The average conclusion time per query was 3.3 seconds, suitable for interactive use on edge GPUs similar as Tesla T4.

Figure 1 llustrates the stationed interface, where the left panel shows stoner queries and uploaded images, and the right panel presents bounding box prognostications and generated answers. The system rightly identifies the scene as a living room with detected objects similar to a lounge, beacon, factory, and television, producing contextually accurate responses.

To validate overall robustness, evaluation was conducted on six indoor test images and 30 questions covering five categories (object listing, room type, counting, attribute, and description).

The bar chart in Fig. **??** demonstrates that all query types achieved near-perfect confidence scores (above 80%), underscoring the SQR module's precision in routing questions to optimal pathways.

Model interpretability was analyzed through attention and density maps that reveal spatial focus regions. Red areas

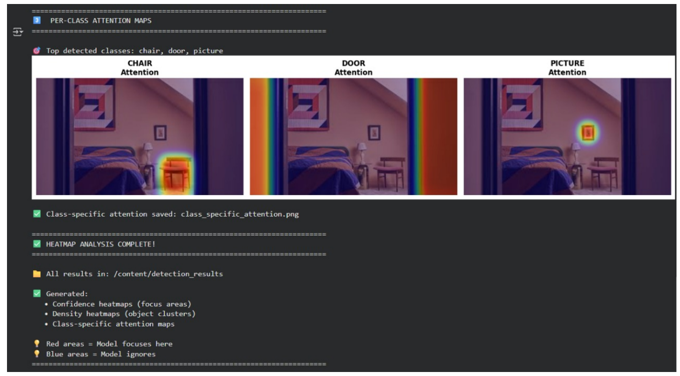

Fig. 3: Attention maps highlighting the regions used for object detection

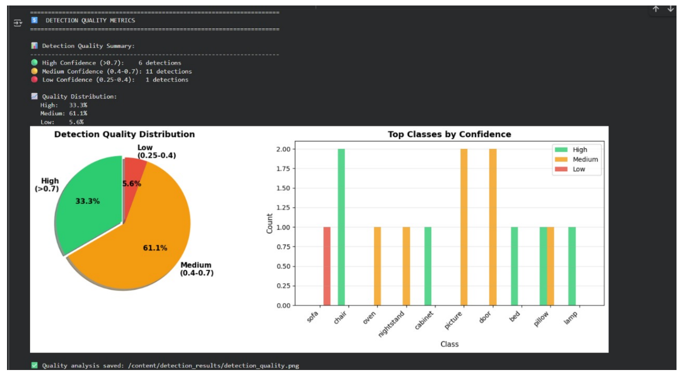

Fig. 4: Confidence distribution of detected objects shown by pie and bar charts

indicate strong model attention, while blue regions are less emphasized.

Attention visualization (Fig. 3) confirms that YOLOv8 focuses precisely on semantic areas such as furniture edges and decor items. The density map (Fig. 2) further verifies the clustering of detected objects in relevant areas (e.g., sofa and cabinet zones).

The detection performance was also evaluated through class-wise confidence analysis and quality distribution visualization.

As seen in Fig. 5, 61.1% of detections had medium confidence (0.4–0.7), while 33.3% were high confidence (> 0.7). This balance ensures robustness across diverse lighting and object-scale variations.

The final evaluation phase compared visual answers generated across different scenes. Examples include detection, routing, and descriptive responses generated by BLIP-2 conditioned on YOLOv8 outputs.

In Fig. 6, the system provides accurate contextual answers, identifying spatial layouts and furnishing details. It demonstrates both object grounding and semantic understanding, bridging the gap between perception and reasoning.

Finally, aggregated metrics and attention-based diagnostics are summarized in Fig. 7. The consistent high performance across detection, routing, and generative reasoning confirms

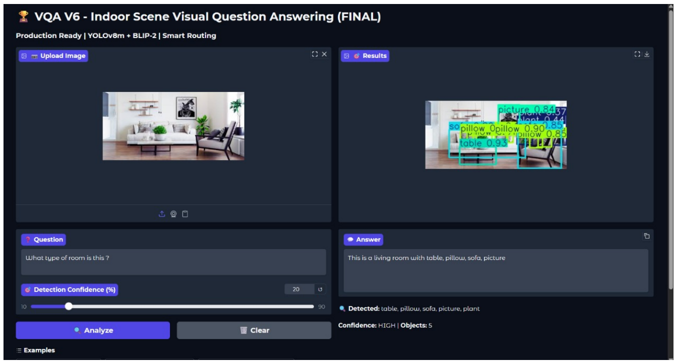

Fig. 5: Distribution of detection confidence across detected classes

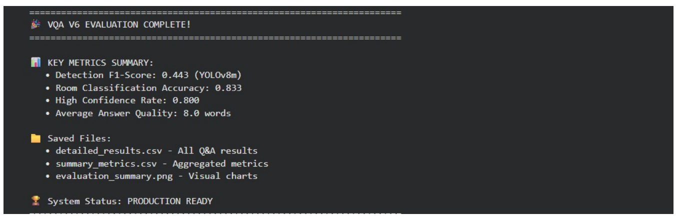

Fig. 6: VQA output showing detected objects and the generated answer

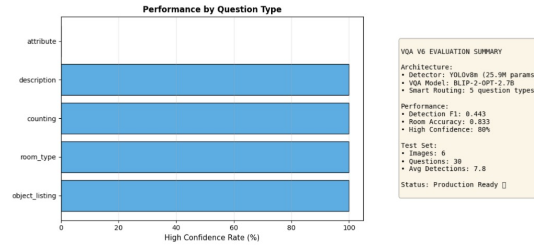

Fig. 7: Summary of detection scores, room accuracy, and answer confidence levels

the robustness of the proposed hybrid VQA system.

Overall, the findings show that basing BLIP-2's logic on YOLOv8 detections significantly lower hallucinations and this increases factual accuracy, which will help us more. The SQR mechanism ensures computational efficiency by delegating simple queries so that the cost is low to deterministic templates. While confidence metrics demonstrate dependability in a variety of indoor environments, attention heatmaps and density visualizations demonstrate spatial interpretability.

We got 0.833 accuracy for room types and about 80% good answers, which basically shows that our system works pretty well in real life.so basically, when we mix the object detection stuff with the language part, it kinda makes the VQA system easier to get and it works very well, like something you can actually use.

VI. Error Analysis and Limitations

Detection misses. Small or occluded objects (e.g., remotes, mugs) are occasionally missed, cascading into VQA errors.

Ambiguous room types. Minimal-context scenes (white walls, single chair) confuse the rule-based classifier.

Open-world terms. Unseen class names in questions (e.g., "ottoman") require BLIP-2 guessing; we mitigate via object-context prompts.

Domain shift. Uncommon décor and lighting reduce confidence; leveraging depth could help.

VII. Ethics and Responsible Use

Indoor perception can reveal sensitive personal information. Our demo processes single images and returns only object names and short answers; no faces or identities are recognized. We recommend on-device processing for home deployments and strict consent for data collection.

VIII. Future Work

3D reasoning: Fuse depth for spatial relations ("behind", "left of").

Temporal memory: Multi-frame tracking for multi-turn dialogue.

Room graph: Scene graphs to reason over affordances and object–room priors.

Edge optimization: INT8/FP8 quantization for Jetson-class devices.

Active learning: Incorporate user feedback into continual training.

IX. Conclusion

Indoor environments are visually dense and semantically rich, which makes them a challenging domain for artificial intelligence. This work presented a unified framework that couples the fast spatial perception of YOLOv8 with the language reasoning capacity of BLIP-2 through a Smart Question Routing (SQR) mechanism, treating detection and question answering as parts of a single pipeline rather than separate problems. Our trials show that a compact sensor, fine- tuned on SUN RGB- D, can give sufficiently strong grounding for a large vision – language model. unequivocal findings help reduce typical visions of generative systems, while BLIP- 2 contributes contextual depth for descriptive or open- concluded questions. Within the estimated setup, the performing system is interpretable and effective enough for interactive use in scripts analogous as assistive interfaces, smart homes, or prototyping for robotics.

The main contribution lies inthe design philosophy separating deterministic perception from probabilistic sense while still passing structured validation between them. This modularity makes the frame easier to extend detectors or VLM backends can be replaced without redesigning the overall architecture. Simple answer cleaning and routing sense further meliorate traceability and perceived responsibility

This study where we have concentrated on static RGB imagery and a limited test set so we can stress test the approach under larger-scale deployment,diffirent house , and more complex thing. Adding depth signals, memory for time, and better ways to reduce data size to the design. would move the system closer to deployment in domestic robots and AR/ VR bias. Overall, our results suggest that orga- nizing living perception and language factors into a rested, modular channel can be as important as gauging individual models. The YOLOv8 – BLIP- 2 combination, guided by SQR, offers one practical step toward inner agents that can both see and speak with better alignment to the visual world

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